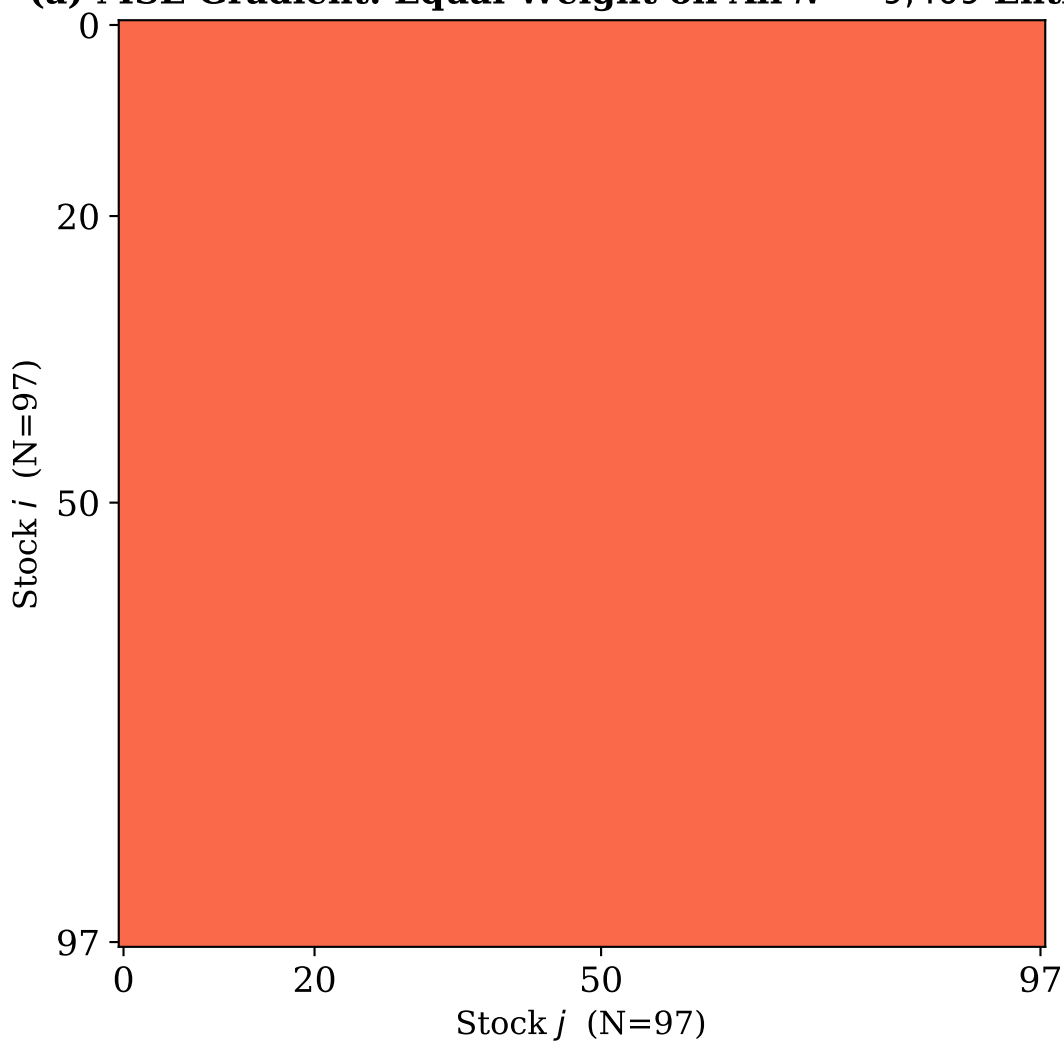
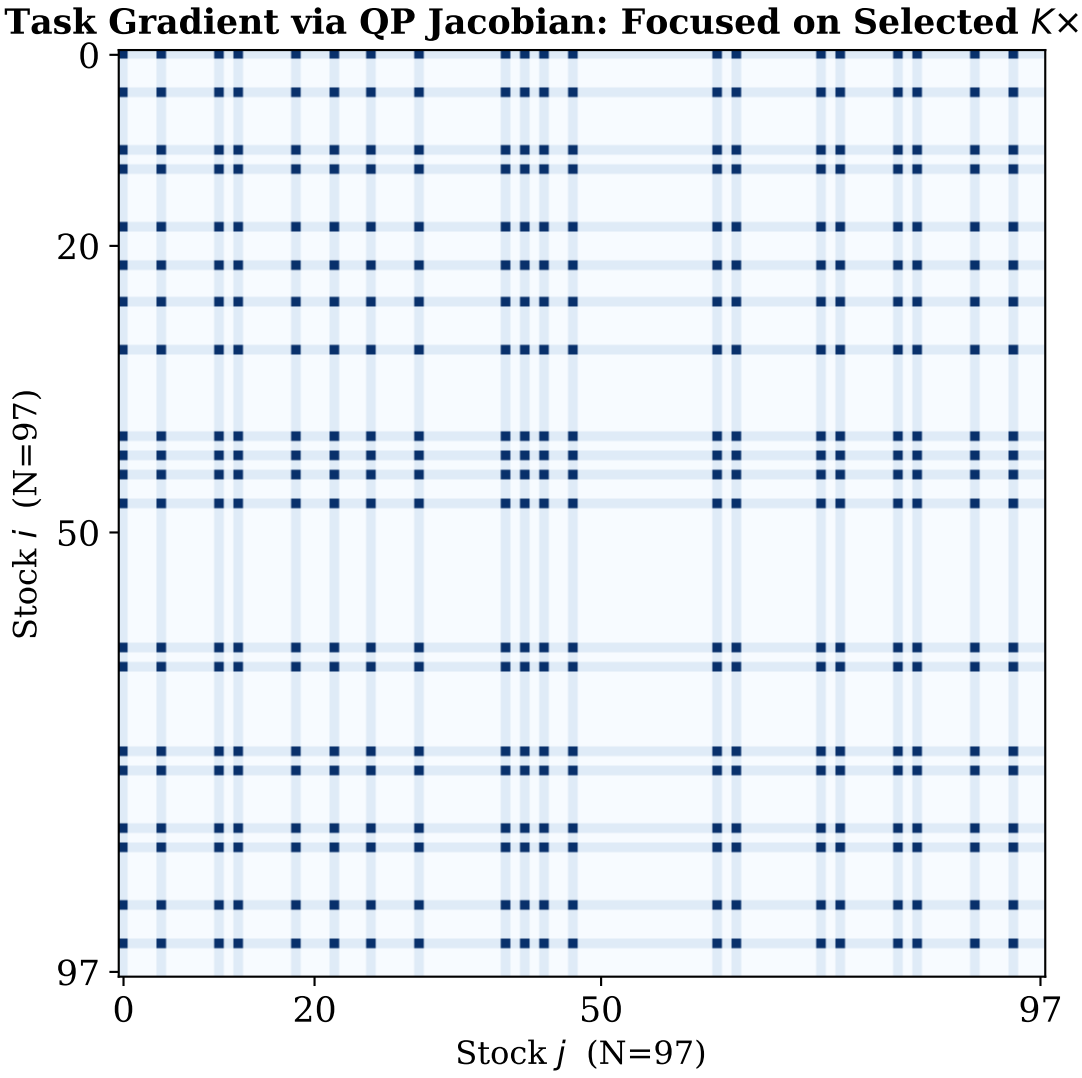
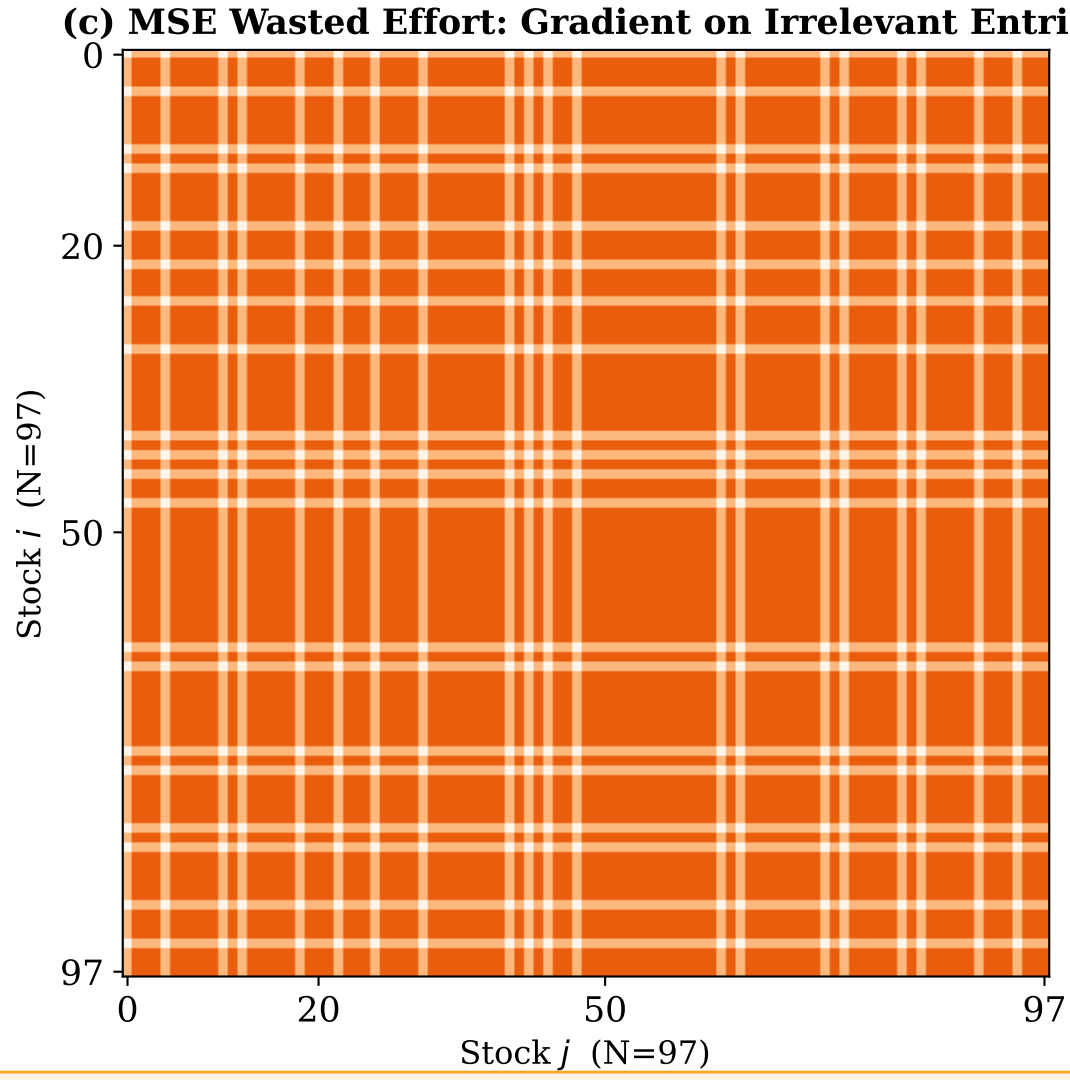
(a) MSE Gradient: Equal Weight on All $N^2 = 9{,}409$ Entries



(b) Task Gradient via QP Jacobian: Focused on Selected $K \times K$ Block



(c) MSE Wasted Effort: Gradient on Irrelevant Entries



At $K/N = 20/97$: only 400 of 9,409 covariance entries matter (4.3%). MSE wastes 95.7% of gradient effort.